

3.1.1 New and emerging technologies

Students must know and understand the impact of new and emerging technologies on contemporary and potential future scenarios in relation to the following areas:

Industry

Content	Potential links to maths and science
The impact of new and emerging technologies on: - the design and organisation of the workplace including automation and the use of robotics - buildings and the place of work - tools and equipment.	

Enterprise

Content	Potential links to maths and science
Enterprise based on the development of an effective business innovation: - crowd funding - virtual marketing and retail - co-operatives - fair trade.	

Sustainability

Content	Potential links to maths and science
The impact of resource consumption on the planet: - finite - non-finite - disposal of waste.	Taking into consideration the ecological and social footprint of materials.

People

Content	Potential links to maths and science
How technology push/market pull affects choice. Changing job roles due to the emergence of new ways of working driven by technological change.	

Culture

Content	Potential links to maths and science
Changes in fashion and trends in relation to new and emergent technologies. Respecting people of different faiths and beliefs.	

Society

Content	Potential links to maths and science
How products are designed and made to avoid having a negative impact on others: • design for disabled • elderly • different religious groups.	

Environment

Content	Potential links to maths and science
Positive and negative impacts new products have on the environment: • continuous improvement • efficient working • pollution • global warming.	

Production techniques and systems

Content	Potential links to maths and science
The contemporary and potential future use of: • automation • computer aided design (CAD) • computer aided manufacture (CAM) • flexible manufacturing systems (FMS) • just in time (JIT) • lean manufacturing.	

How the critical evaluation of new and emerging technologies informs design decisions

Content	Potential links to maths and science
That it is important to consider scenarios from different perspectives and considering: • planned obsolescence • design for maintenance • ethics • the environment.	Ethical factors and consideration of ecological and social footprint.